

Copilot Project Portfolio

Technical case studies · Sanitized reference patterns

This document summarizes representative Microsoft Copilot and enterprise AI delivery patterns. Diagrams and deeper notes live in the companion GitHub repository **AI-Copilot-Portfolio**.

label: Sanitized Reference Architecture — not proprietary client code

01 — Enterprise Knowledge Copilot

Business problem. Employees struggle to find verified policies and procedures across SharePoint and approved knowledge stores.

Solution. Copilot Studio assistant with RAG over enterprise documents; Azure OpenAI grounded generation with citations.

Technologies. Copilot Studio, Azure OpenAI, embeddings, vector search, SharePoint, Dataverse, Entra ID.

Security. Authorization-aware retrieval, RBAC, managed identity, Purview-aligned classification, audit of retrieval IDs.

Value. Faster, consistent answers tied to sources employees are entitled to open.

02 — Process Automation Agent

Business problem. Routine service requests and validations create backlog and inconsistent data entry.

Solution. Intent detection in Copilot Studio → Power Automate / REST → Dataverse or backend → structured status.

Technologies. Copilot Studio, Power Automate, custom connectors, OAuth 2.0, Dataverse, REST APIs.

Security. Least-privilege scopes, error handling without leaking internals, approvals for high-impact writes.

Value. Shorter cycle times; example adjacent automation reduced file validation from ~3 days to ~10 minutes.

03 — Agentic AI Workflow

Business problem. Users ask mixed questions requiring knowledge, data, APIs, or workflows—scripted topics do not scale.

Solution. Orchestrator selects tools: RAG, structured data, REST/MCP, Power Automate, specialist agents, or refusal.

Technologies. Azure OpenAI tool calling, Copilot Studio, MCP/API integrations, SharePoint, Dataverse, Databricks.

Security. Narrow tool permissions; human approval for high-risk writes; full tool-invocation logging.

Value. One controlled entry point for knowledge and actions without entitlement expansion.

04 — Enterprise RAG Architecture

Business problem. Ungrounded generation is unsuitable for regulated Q&A.;

Solution. Ingest → preprocess → chunk → embed → vector index → filtered retrieve → prompt assembly → evaluate.

Technologies. Azure OpenAI embeddings/completions, vector index, metadata filters, evaluation datasets.

Security. ACL-aware retrieval, prompt-injection isolation, sensitivity tiers, release gates on groundedness.

Value. Defensible answers with measurable quality and latency.

05 — Copilot + Microsoft Fabric

Business problem. AI over raw tables yields conflicting metrics and unclear ownership.

Solution. Fabric Lakehouse Bronze/Silver/Gold → semantic model → Power BI and Copilot/downstream AI.

Technologies. Fabric, Dataflows/pipelines, ADF/Databricks where applicable, Power BI, SQL/Python, Purview.

Security. Workspace RBAC, RLS on models, AI-approved entities documented separately from raw landing zones.

Value. Shared business definitions across reports and conversational AI.

06 — AI Security & Governance

Business problem. Agents can become unintended entitlement bypasses without deliberate controls.

Solution. Framework covering Entra ID, RBAC/RLS, managed identity, Key Vault, DLP, Purview, Responsible AI, env isolation.

Technologies. Entra ID, Power Platform DLP, Purview, Key Vault, audit logging, OAuth.

Security. Principle: never exceed the authenticated user's normal access for information or operations.

Value. Audit-ready AI delivery aligned to existing IAM programs.

07 — Copilot ALM / Production Deployment

Business problem. Portal-only development lacks review, regression, and rollback discipline.

Solution. Dev → source control → Test → UAT → Prod using managed solutions, env vars, connection references, CI/CD.

Technologies. Power Platform ALM, Azure DevOps or GitHub Actions, monitoring and adoption analytics.

Security. Least-privilege deploy identities; secrets outside the repository; approval gates for Prod.

Value. Predictable releases with evidence for regulated change management.

Architecture Pattern (Knowledge Copilot)

Employee → Entra ID → Copilot Studio → orchestration / grounding → RAG retrieval (embeddings, vector search, ACL/metadata filters) → Azure OpenAI → grounded response with citations. Knowledge sources typically include SharePoint, approved document sets, and selected Dataverse content.

Architecture Pattern (Agentic Action)

User intent → orchestrator → choose among knowledge search, structured data read, REST/MCP action, Power Automate workflow, specialist agent, or refusal. High-risk writes pause for human approval. Every tool call is schema-validated and audited.

Implementation Considerations

- Start with grounded Q&A; and measurement before enabling write tools.
- Prefer curated Fabric Gold / semantic models for analytical answers.
- Separate read and write identities; require human approval for high-risk actions.
- Treat retrieved document text as untrusted for instruction override.
- Promote with evaluation evidence and managed solutions—not manual portal edits in Prod.

Agent Capabilities Summary

Capability	Pattern
Knowledge Q&A;	RAG + citations + refusal
Status lookup	Authorized API / Dataverse read
Request intake	Validated create via flow/API
Approvals	Human-in-the-loop on sensitive writes
Delegation	Specialist agents for narrow domains
Analytics answers	Governed semantic model / Gold data

Delivery & ALM Snapshot

Solutions are developed in Dev as unmanaged packages, stored in source control, and promoted as managed solutions through Test, UAT, and Production. Environment variables and connection references keep environment-specific values out of hard-coded topics. CI/CD (Azure DevOps or GitHub Actions) applies approval gates; AI evaluation regression sets protect groundedness and relevance before production changes.

Representative Experience Context

Regulatory / insurance-style programs: Copilot Studio, Azure AI/OpenAI, Power Platform, Dataverse, SharePoint, RAG, embeddings, vector search, prompt engineering, agentic workflows, connectors, Responsible AI, RBAC, AI evaluation, ALM/CI/CD, Purview, Entra ID, Fabric Dataflows/Lakehouse, Power Automate.

Banking-style programs: Copilot Studio assistants, Azure AI/OpenAI, agentic workflows, SharePoint, SAP integration layers, Databricks, Dataverse, MCP/API integrations, RAG pipelines, evaluation, security/governance, ALM/DevOps.

Fabric / analytics: Bronze/Silver/Gold Lakehouse, Fabric, ADF, Synapse, Databricks, Power BI, semantic models, SQL, Python, Power Automate.

Security Controls Applied Across Projects

Control	How it shows up in delivery
Entra ID	Every user session authenticated; groups drive AuthZ filters
RBAC / RLS	Tool access and semantic-model rows scoped to role
Managed identity	Azure resources avoid embedded secrets where possible
OAuth scopes	Read vs write API permissions separated
Key Vault	Remaining secrets outside source control and topics
DLP / connectors	Approved connector set in Power Platform
Purview	Classification influences what may enter indexes
Audit	Correlation IDs on retrieval and tool calls
Environments	Dev / Test / UAT / Prod isolation
Human approval	Required before high-risk write tools commit

Outcomes Stakeholders Typically Track

- Time-to-answer for policy and procedure questions
- Citation coverage rate on knowledge responses
- Automation rate for routine service requests
- Evaluation scores (groundedness / relevance) on release candidates
- Flow failure rate and mean time to recover after a bad deploy
- Adoption and containment metrics from Copilot analytics

Confidentiality: Client-specific implementations, source code, datasets, credentials, internal URLs, and proprietary architecture details are intentionally excluded.